Almost everywhere divergence of polynomial interpolation with sublinear excess degree

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Abstract

Let X_n be an arbitrary set of n distinct points in $[-1,1]$, and let r_n be nonnegative integers with $r_n = o(n)$. We prove that there is a real continuous function f such that every sequence of polynomials p_n, of degree at most $n+r_n$, satisfying $p_n = f$ on X_n, has $\limsup_{n\to\infty} |p_n(x)| = \infty$ for almost every $x \in [-1,1]$. The exceptional null set may depend on the sequence (p_n). This extends the Erdős–Vértesi theorem on almost-everywhere unboundedness of Lagrange interpolation for arbitrary node arrays to nonunique interpolation with $o(n)$ excess degree, and in particular answers Erdős Problem 1152 in a stronger form. The proof combines localized functions in Bernstein spaces with uniform asymptotics for weighted Christoffel–Darboux kernels. A finite interpolation construction and the Baire category theorem give a single function valid for every admissible interpolation sequence.

1 Introduction

Write Π_d for the space of real polynomials of degree at most d. All measures of subsets of $[-1,1]$ below are Lebesgue measures. The uniform norm on this interval is denoted by $\|\cdot\|_\infty$.

Theorem 1. *For each $n \geq 1$, let $X_n \subset [-1,1]$ consist of n distinct points. Suppose that $r_n \geq 0$ are integers and $r_n/n \to 0$. There exists $f \in C([-1,1];\mathbb{R})$ such that every sequence satisfying*

$$p_n \in \Pi_{n+r_n}, \qquad p_n(x) = f(x) \quad (x \in X_n)$$

also satisfies

$$\limsup_{n\to\infty} |p_n(x)| = \infty \quad \textit{for almost every } x \in [-1,1]. \tag{1.1}$$

Erdős and Vértesi proved that for every triangular array of distinct nodes in $[-1,1]$, there is a continuous function whose Lagrange interpolation polynomials are unbounded almost everywhere [1, 2]. Theorem 1 extends this arbitrary-node result to nonunique interpolation: degrees up to $n+r_n$, with $r_n = o(n)$, are allowed, and the same continuous function works for every admissible sequence of interpolants. This also answers Erdős Problem 1152 [4] in the stronger form (1.1). The exceptional null set may depend on the sequence (p_n).

The contrast with fixed positive excess degree is provided by Erdős, Kroó, and Szabados [3]: for suitable node arrays and each fixed $\varepsilon > 0$, every continuous function admits uniformly convergent interpolants of degree at most $\lfloor (1+\varepsilon)n \rfloor$. For background on divergence phenomena in Lagrange interpolation, see Szabados and Vértesi [5]. The same conclusion as Theorem 1 holds for polynomials with complex coefficients, by applying the real statement to their real parts.

The two analytic ingredients are the construction in Olevskii and Ulanovskii [7, Lemma 4.1] and the uniform kernel asymptotics of Kriecherbauer, Schubert, Schüler, and Venker [6, Theorem 1.3 and Theorem 1.8 (bulk universality)]. Section 3 records the kernel estimates in the normalization used below, while Section 6 derives the corresponding local external field from a weak limit of the node measures. We first obtain interpolation data that force large values on a fixed fraction of a local interval, uniformly over all admissible polynomial corrections. We then combine finitely many such data sets to rule out common sets of bounded values. The final step is a category argument in $C([-1,1];\mathbb{R})$.

Comments and corrections are welcome.

2 Localized Bernstein functions and polynomial corrections

For $\sigma > 0$, let B_σ denote the space of entire functions of exponential type at most σ that are bounded on $\mathbb{R}$. Norms of functions in B_σ are taken on $\mathbb{R}$.

Lemma 2. *There are absolute constants $c, C > 0$ with the following property. For every sufficiently large integer ρ, put*

$$R_0 = \rho - \sqrt{\rho}, \qquad \gamma = R_0^{-1/9}.$$

For every finite set $\Gamma \subset [-\rho, \rho)$ with $\#\Gamma \le 2\rho + 1$, there is an entire function g, real on $\mathbb{R}$, such that

$$g \in B_{\pi-4\gamma}, \qquad \max_\Gamma |g| \le 1, \qquad \|g\|_\infty \le C \log \rho, \tag{2.1}$$

$$g(u_0) \ge c \log \rho \quad \textit{for some } |u_0| \le R_0, \tag{2.2}$$

$$|g(u)| \le C(|u| - R_0)^{-3} \quad (|u| \ge \rho). \tag{2.3}$$

These functions have uniformly bounded $L^1(\mathbb{R})$ norms and uniformly vanishing integral tails when ρ is fixed. There is also an interval, of length bounded below by an absolute positive constant, contained in $(-\rho, \rho) \setminus \Gamma$, on which $g \ge c' \log \rho$, with an absolute constant $c' > 0$.

Proof. Adapt the construction of [7, Lemma 4.1, pp. 184–185] by reducing its initial bandwidth from $\pi - 4R_0^{-1/9}$ to $\pi - 8R_0^{-1/9}$. The fourth-power sinc factor below then leaves the required margin 4γ below π. Put $w = \pi - 8\gamma$ and

$$\Lambda = \Gamma + 2R_0\mathbb{Z}.$$

Since Γ is finite, Λ is uniformly discrete. Being periodic, its lower and upper uniform densities coincide; write their common value as D. Then

$$D \le \frac{2\rho + 1}{2R_0} \le 1 + CR_0^{-1/2}.$$

Write $K_s(\Lambda, B_w)$ for the sampling constant. If Λ is a sampling set for B_w, then [7, Theorem C(i), p. 179] gives $D > w/\pi$. Hence $D\Lambda$ has density one and [7, Theorem 1(i), p. 180] gives

$$K_s(\Lambda, B_w) = K_s(D\Lambda, B_{w/D}) \ge C \log \frac{\pi}{\pi - w/D} \ge c \log R_0,$$

since, for large ρ,

$$0 < \pi - \frac{w}{D} = \frac{\pi D - w}{D} \le C(R_0^{-1/2} + \gamma) \le C\gamma.$$

If Λ is not a sampling set, its sampling constant is infinite, so the same lower bound is immediate. Thus one can choose $g_1 \in B_w$, bounded by one on Λ, whose norm is a fixed constant times $\log R_0$. Choose a point at which its absolute value is at least half this norm. A translation by a multiple of $2R_0$ places this point at $u_0 \in [-R_0, R_0]$, without changing the bound on Λ. Multiply by a constant of modulus one so that $g_1(u_0) > 0$, then replace $g_1(z)$ by $(g_1(z) + \overline{g_1(\bar{z})})/2$. This entire function is real on $\mathbb{R}$, retains the value at u_0, and satisfies the same bandwidth and upper bounds.

Multiply this function by

$$\left(\frac{\sin(\gamma(u-u_0))}{\gamma(u-u_0)}\right)^4.$$

The resulting function has bandwidth at most $\pi - 4\gamma$ and retains the node bound and the value at u_0. For $|u| \geq \rho$, the bound

$$C \log R_0\, \gamma^{-4}(|u| - R_0)^{-4} \leq C(|u| - R_0)^{-3}$$

follows from $\log R_0\, R_0^{4/9}/\sqrt{\rho} = O(1)$. This proves (2.1)–(2.3) and the assertions about integrability. Finally, Bernstein's inequality gives $\|g'\|_\infty \leq \pi C \log \rho$. Thus the peak contains an interval of fixed positive length on which $g \geq c' \log \rho > 1$. This interval contains no point of Γ and lies inside the original box. $\square$

For a finite set $Y \subset \mathbb{R}$, define

$$P_Y(x) = \prod_{y \in Y} (x - y).$$

Lemma 3. *Let $J_1, \ldots, J_K$ be disjoint intervals, in increasing order, none of which meets Y. Suppose that a real polynomial b satisfies*

$$|b| > H, \qquad \operatorname{sgn}(b/P_Y)|_{J_j} = (-1)^j. \tag{2.4}$$

If $p = b + P_Y q$ with $\deg q \leq d$, then at least $(K - d - 1)/2$ of these intervals satisfy $|p| > H$ at every point.[1]

Proof. Call J_j a low interval if it contains a point x_j with $|p(x_j)| \leq H$. At such a point the sign of $q(x_j)$ is opposite to that of $b(x_j)/P_Y(x_j)$. If there are B low intervals, the resulting signs alternate at least $2B - K - 1$ times. Indeed, deleting $K - B$ terms from a fully alternating sequence removes at most $2(K - B)$ sign changes. Consequently $2B - K - 1 \leq d$, so

$$B \leq \frac{K + d + 1}{2}. \tag{2.5}$$

The asserted bound follows. $\square$

3 An explicit external field and its polynomial kernel

For a finite positive measure μ, use the logarithmic-potential convention

$$V_\mu(x) = \int \log |x - y| \, d\mu(y).$$

Define

$$Q(u) = \frac{1}{2}\big((1+u)\log(1+u) + (1-u)\log(1-u)\big), \qquad |u| < 1. \tag{3.1}$$

Then $Q(0) = 0$, $Q'(u) = \operatorname{arctanh} u$, and $Q''(u) = (1 - u^2)^{-1}$.

[1] This mechanism was developed in an earlier GPT-5.6-assisted round of the project.

Lemma 4. *For $0 < t < 1$, set $a_t = \sqrt{1-(1-t)^2}$. The equilibrium measure of mass t in the external field Q on $[-1,1]$ has density*

$$\rho_t(u) = \frac{1}{\pi} \arccos \frac{1-t}{\sqrt{1-u^2}} \, \mathbf{1}_{\{|u|<a_t\}}. \tag{3.2}$$

Thus $Q - V_{\rho_t}$ is constant on $[-a_t, a_t]$ and is no smaller elsewhere in $[-1,1]$.

Proof. Differentiation with respect to t, away from $|u| = a_t$, gives

$$\partial_t \rho_t(u) = \frac{\mathbf{1}_{\{|u|<a_t\}}}{\pi\sqrt{a_t^2 - u^2}}. \tag{3.3}$$

This is the arcsine probability density on $[-a_t, a_t]$, so ρ_t has total mass t. Its Cauchy transform can be computed by integrating the transforms of these arcsine measures. For $0 < x < a_t$, its real boundary value is

$$\int_0^{1-\sqrt{1-x^2}} \frac{ds}{\sqrt{x^2 - a_s^2}} = \operatorname{arctanh} x = Q'(x). \tag{3.4}$$

One may justify the calculation first in the upper half-plane by Fubini's theorem and then take boundary values. For $a_t < x < 1$, the upper limit in (3.4) is t, and the integral is strictly smaller than $Q'(x)$. Symmetry gives the corresponding statements on the negative half-axis. These identities prove the equilibrium equalities and inequalities. $\square$

Put $A = [-1/4, 1/4]$. We will use the following consequences of [6].

Lemma 5 (Uniform weighted kernel estimates)**.** *Fix $0 < t < 1$ sufficiently close to one that $A \Subset (-a_t, a_t)$, and choose $a_t < b < 1$. There are intervals*

$$A \Subset J_0 \Subset J_1 \Subset (-b, b)$$

and an analytic neighborhood of Q on a fixed complex neighborhood of $[-b,b]$ with the following property. Let $s \to \infty$, $m = \lfloor ts \rfloor$, and let Q_s belong to this neighborhood. If $p_0, \ldots, p_{m-1}$ are orthonormal for $e^{-2sQ_s(x)}\,dx$ on $[-b,b]$, set

$$K_s(x,z) = e^{-sQ_s(x)-sQ_s(z)} \sum_{j=0}^{m-1} p_j(x)p_j(z), \qquad c_s = \frac{s}{2}. \tag{3.5}$$

Let $\beta_{s,y}$ be twice the density at y of the equilibrium measure of mass m/s for Q_s. Then, uniformly for the indicated family of external fields,

$$\frac{K_s(y+u/c_s, y+v/c_s)}{c_s} = \frac{\sin(\pi\beta_{s,y}(u-v))}{\pi(u-v)} + o(1), \tag{3.6}$$

$$\frac{|K_s(x,z)|}{c_s} \le \frac{C}{1+s|x-z|} \qquad (x,z \in J_1), \tag{3.7}$$

$$\frac{|K_s(x,z)|}{c_s} \le Cs^{-5/6} \qquad (x \in [-b,b] \setminus J_0,\ z \in A + O(s^{-1})). \tag{3.8}$$

In (3.6), $y \in A$, u, v range over bounded sets, and the quotient at $u = v$ is interpreted by continuity.

Proof. Set

$$V_s = \frac{2s}{m} Q_s,$$

so that $e^{-mV_s} = e^{-2sQ_s}$. The limiting field $2Q/t$ is analytic and strictly convex, and its equilibrium support is contained in the interior of $[-b, b]$; after shrinking the neighborhood of Q, the same regular one-cut situation holds uniformly for V_s.

Let ψ_s be the equilibrium probability density for V_s. Then $\rho_s = (m/s)\psi_s$ is the equilibrium density of mass m/s for Q_s, and

$$m\psi_s(y) = s\rho_s(y) = c_s \beta_{s,y}.$$

Substituting the local variables $\beta_{s,y}u$ and $\beta_{s,y}v$ in [6, Theorem 1.8 (bulk universality)] gives (3.6); the uniformity follows from part (b) of that theorem.

For (3.7), use the difference-quotient formula in [6, Theorem 1.3]. Near the diagonal the same estimate follows from the diagonal bound and Cauchy–Schwarz. For (3.8), the Airy formulas in that theorem give an $O(s^{1/6})$ factor at a support endpoint, while the factor at $z \in A + O(s^{-1})$ remains bounded; since x and z are separated by a fixed positive distance and $c_s = s/2$, the stated $s^{-5/6}$ bound follows. The estimates are uniform after the external-field neighborhood is fixed. □

4 Local amplification with the external-node weight

Consider a sequence of finite node sets $Y \subset [-1, 1]$, with $N = \#Y$, and fix

$$I = (x_0 - h, x_0 + h) \Subset (-1, 1), \qquad s = \#(Y \cap I).$$

In the coordinate $u = (x - x_0)/h$, define

$$W(u) = \prod_{y \in Y \setminus I} (x_0 + hu - y), \qquad Q_s(u) = -\frac{1}{s} \log \left| \frac{W(u)}{W(0)} \right|. \tag{4.1}$$

The function Q_s is convex on $(-1, 1)$.

Proposition 6. *Fix $H > 1$, and choose the parameters of Sections 2 and 3 as below. Suppose that $s \to \infty$, that Q_s lies in the specified analytic neighborhood of Q, and that*

$$\#\{y \in Y : (y - x_0)/h \in A\} \leq \frac{s}{4} + \epsilon_\rho s, \qquad \epsilon_\rho = \frac{1}{16\rho}. \tag{4.2}$$

If $d = o(s)$, then, for all sufficiently large s, there are data $v_y \in [-1, 1]$, $y \in Y$, such that every $p \in \Pi_{N+d}$ interpolating these data satisfies

$$|\{x \in I : |p(x)| > H\}| \geq c_H |I|. \tag{4.3}$$

The constant $c_H > 0$ depends only on H.

Proof. We fix the auxiliary parameters in the following order, all before letting $s \to \infty$. First choose ρ so large that

$$c' \log \rho > 10H,$$

where c' is the constant in Lemma 2, and set $R_0 = \rho - \sqrt{\rho}$ and $\gamma = R_0^{-1/9}$. Choose $t < 1$ so close to one that

$$\inf_{y \in A} 2\pi\rho_t(y) > \pi - 2\gamma.$$

Fix $K = 3$, and then choose $\kappa > 0$ so small that $t + \kappa < 1$ and the Jackson factor (4.8) below is larger than $9/10$ whenever $y \in A$ and $|c_s(x - y)| \le \rho + 1$, for all sufficiently large s. Choose b with $a_{t+\kappa} < b < 1$, choose $A \Subset J_0 \Subset J_1 \Subset (-b, b)$ inside the common bulk region, and shrink the analytic neighborhood of Q so that the kernel estimates of Section 3 are uniform there, the supports of the equilibrium measures of masses between t and $t + \kappa$ stay a fixed positive distance from $\pm b$, and

$$\inf_{y \in A} \pi \beta_{s,y} > \pi - 3\gamma. \tag{4.4}$$

Set $\eta = 1/10$. By the uniform integral-tail bound in Lemma 2, choose T so that the truncation error in (4.6) is less than $\eta/2$. Next choose $R > \max\{\rho + 1, 2T\}$. Finally choose $D > 2R$ so that

$$2C_\rho \sum_{j=0}^{\infty} \frac{1}{1 + R + jD} \frac{1}{(1 + \kappa(R + jD))^{2K}} < \frac{1}{10}. \tag{4.5}$$

Here C_ρ is the constant in (4.10); first increasing R and then D makes (4.5) hold. All these parameters depend only on H. In this proof, the arguments of K_s, $G_{s,y}$, and $F_{s,y}$ are in the scaled coordinate u.

Partition the interval $c_s A$ into boxes of length 2ρ, discarding incomplete boxes at the ends. There are $s/(8\rho) - O(1)$ boxes. Call a box admissible if it contains at most $2\rho + 1$ scaled nodes. If B is the number of admissible boxes, then

$$\left(\frac{s}{8\rho} - O(1) - B \right) (2\rho + 2) \le \frac{s}{4} + \epsilon_\rho s.$$

The choice of ϵ_ρ implies $B \ge c_\rho s$.

For an admissible box centered at $c_s y$, apply Lemma 2 to its translated nodes, obtaining g. For a fixed truncation parameter T, define

$$G_{s,y}(x) = \int_{-T}^{T} g(v) \frac{K_s(x, y + v/c_s)}{c_s} \, dv. \tag{4.6}$$

This is $e^{-sQ_s(x)}$ times a polynomial of degree at most $m - 1$. By (4.4) and the Paley–Wiener theorem,

$$\int_{\mathbb{R}} g(v) \frac{\sin(\pi \beta_{s,y}(u - v))}{\pi(u - v)} \, dv = g(u).$$

With η, T, R fixed above, Equation (3.6) implies, for all sufficiently large s,

$$\sup_{|u| \le R} |G_{s,y}(y + u/c_s) - g(u)| < \eta. \tag{4.7}$$

The uniform L^1 bound makes this assertion uniform even when g varies with the node configuration.

To obtain summable tails, multiply by an algebraic Jackson factor. Write $x = \cos\theta$, $y = \cos\alpha$, and, with $K = 3$ and κ fixed above, set $k = \lfloor \kappa s/(2K) \rfloor$. Define

$$\begin{aligned} D_k(v) &= \frac{\sin((k + 1/2)v)}{(2k + 1)\sin(v/2)}, \\ L_{s,y}(x) &= D_k(\theta - \alpha)^{2K} + D_k(\theta + \alpha)^{2K}. \end{aligned} \tag{4.8}$$

This expression is even in θ, so it is an algebraic polynomial in x of degree at most $2Kk$. Since $y \in A$, it is bounded on $[-1, 1]$ by $1 + o(1)$. By the choice of κ, it is greater than $9/10$ throughout the scaled neighborhood containing the peak interval of g. Moreover,

$$|L_{s,y}(x)| \le C(1 + \kappa s|x - y|)^{-2K} + Cs^{-2K}. \tag{4.9}$$

Put $F_{s,y} = G_{s,y} L_{s,y}$. For $x \in J_1$ with $|c_s(x-y)| > 2T$, (3.7) gives

$$\begin{aligned} |F_{s,y}(x)| \le C_\rho (1 + |c_s(x-y)|)^{-1} (1 + \kappa |c_s(x-y)|)^{-2K} \\ + O_\rho(s^{-2K}). \end{aligned} \tag{4.10}$$

For $x \in [-b,b] \setminus J_0$, (3.8) and (4.9) give

$$|F_{s,y}(x)| = O_\rho(s^{-2K-5/6}). \tag{4.11}$$

To treat $b < |x| < 1$, write

$$F_{s,y}(x) = e^{-sQ_s(x)} H_{s,y}(x), \qquad \deg H_{s,y} \le D_s := m - 1 + 2Kk < s.$$

Let σ be the equilibrium measure of mass D_s/s for the field Q_s on $[-b,b]$. Its support remains in a fixed compact subinterval of $(-b,b)$. This follows from the continuity of the support endpoints in [6, Lemma 2.4], after shrinking the external-field neighborhood to cover masses near both t and $t+\kappa$. Let F be the constant value of $Q_s - V_\sigma$ on $\operatorname{supp}\sigma$. Since $\deg H_{s,y} \le D_s = s\sigma(\mathbb{R})$, the maximum principle applied to $\log|H_{s,y}| - sV_\sigma$ on the complement of the support, including infinity, yields

$$|e^{-sQ_s(x)} H_{s,y}(x)| \le \left\| e^{-sQ_s} H_{s,y} \right\|_{\operatorname{supp}\sigma} e^{-s(Q_s(x) - V_\sigma(x) - F)}. \tag{4.12}$$

For $x \in \operatorname{supp}\sigma \cap J_0$, the points $y + v/c_s$, $|v| \le T$, lie in J_1 for all large s. Hence (3.7), the uniform L^1-bound on g, and $\|L_{s,y}\|_\infty \le 1 + o(1)$ give

$$|F_{s,y}(x)| \le C_\rho.$$

For $x \in \operatorname{supp}\sigma \setminus J_0$, the same conclusion follows from (3.8), since $y + v/c_s$ stays within $O(s^{-1})$ of A. Consequently

$$\left\| e^{-sQ_s} H_{s,y} \right\|_{\operatorname{supp}\sigma} = \|F_{s,y}\|_{\operatorname{supp}\sigma} \le C_\rho \tag{4.13}$$

uniformly in the admissible box and in the allowed external field.

Put $D_s^*(x) = Q_s(x) - V_\sigma(x) - F$. The function Q_s is convex on $(-1,1)$, while $V_\sigma'' < 0$ off $\operatorname{supp}\sigma$. After shrinking the analytic neighborhood of Q, there is $c_0 > 0$ such that $Q_s'' \ge c_0$ on $[-b,b]$, and hence

$$(D_s^*)'' = Q_s'' - V_\sigma'' \ge c_0 \qquad \text{off } \operatorname{supp}\sigma \text{ in } [-b,b].$$

Let α_s and β_s be the leftmost and rightmost points of $\operatorname{supp}\sigma$. The support remains a fixed positive distance from $\pm b$. Since $D_s^* \ge 0$ on $[-b,b]$ and $D_s^* = 0$ on the support, its outward one-sided derivative at α_s and β_s is nonnegative. Integrating the preceding inequality from the support to $\pm b$ gives, for some $c > 0$ independent of s,

$$D_s^*(-b) \ge c, \qquad D_s^*(b) \ge c.$$

Outside $[-b,b]$, strict convexity on the two complementary intervals shows that D_s^* continues to increase toward the endpoints. Therefore (4.12) and (4.13) give

$$|F_{s,y}(x)| = O_\rho(e^{-cs}) \qquad (b < |x| < 1). \tag{4.14}$$

Retain admissible boxes whose centers in the $c_s u$ coordinate are separated by at least the fixed distance $D > 2R$. By (4.5), the sum of the tails in (4.10) from all centers outside the R-window is less than $1/10$. One residue class of the box indices retains at least $c_{\rho,D} s$ admissible boxes. At any point there is at most one center within scaled distance R. In this window use (4.7). At nodes of the same box use $|g| \le 1$; outside the box use (2.3). The remaining terms are controlled by the absolutely summable tails (4.10)–(4.14). The $O_\rho(s^{-2K})$ remainders remain $o(1)$ after summing over all retained boxes. Thus, for every choice $\varepsilon_j \in \{-1, 1\}$,

$$\max_{z \in Y \cap I} \left| \sum_j \varepsilon_j F_{s,y_j} \left(\frac{z - x_0}{h} \right) \right| \le 2. \tag{4.15}$$

On $(-1, 1)$, the ratio $W(u)/W(0)$ is positive, and hence $e^{-sQ_s(u)} = W(u)/W(0)$. The sum in (4.15) therefore extends to a polynomial in the original variable, vanishes at $Y \setminus I$, and has degree at most

$$N - s + m - 1 + 2Kk \le N - 1. \tag{4.16}$$

Divide it by two. Its values at all nodes now belong to $[-1, 1]$. On the peak interval belonging to the j-th center, (4.7), the choice of κ, and $c' \log \rho > 10H$ give

$$|F_{s,y_j}| \ge \frac{9}{10}(10H - \eta).$$

The other retained centers contribute at most $1/10 + o(1)$ there by (4.5) and (4.10)–(4.14). Hence, for all sufficiently large s, each retained peak contains a node-free interval of length c/s in the u-coordinate on which the polynomial after division by two has absolute value greater than $2H$, with sign determined by ε_j. Choose these signs so that the resulting polynomial, denoted by b, satisfies (2.4). Set $v_y = b(y)$. Every interpolant $p \in \Pi_{N+d}$ has the form $p = b + P_Y q$, with $\deg q \le d$. There are at least $c_{\rho,D} s$ peak intervals, whereas $d = o(s)$. Lemma 3 preserves a fixed fraction of them as intervals on which $|p| > H$ everywhere. Returning to the original variable proves (4.3). $\square$

5 Intervals with small node density

A second local construction requires no external field. Let $I \Subset (-1, 1)$, let Y contain N nodes, and write

$$x = \cos\theta, \qquad v = \frac{N\theta}{\pi}.$$

Let $I^\theta \subset (0, \pi)$ be the angular interval corresponding to I. Suppose that

$$\#(Y \cap I) \le \frac{N|I^\theta|}{2\pi}. \tag{5.1}$$

Choose ρ so that $c' \log \rho > 10H$, and then choose $D > 4\rho$ so that

$$2C \sum_{j=1}^{\infty} (jD - 2\rho)^{-3} < \frac{1}{10}. \tag{5.2}$$

Partition the corresponding v-interval into boxes of length 2ρ, discarding the incomplete boxes at the two ends. Let M be the number of complete boxes and B_{bad} the number containing at least $2\rho + 2$ nodes. The v-interval has length $L = N|I^\theta|/\pi$, so (5.1) gives

$$B_{\text{bad}}(2\rho + 2) \le \frac{L}{2}, \qquad L < 2\rho(M + 2).$$

Thus

$$B_{\text{bad}} < \frac{\rho(M+2)}{2\rho+2} \le \frac{M}{2} + 1.$$

Hence at least $M/2 - 1$ complete boxes contain at most $2\rho + 1$ nodes; in particular, this is at least $M/3$ once $M \ge 6$. Retain such boxes with a fixed separation D, and apply Lemma 2 at their centers a_j. For the resulting functions g_j, define

$$\Phi_{N,j}(\theta) = \sum_{\ell \in \mathbb{Z}} \left[g_j\left(\frac{N\theta}{\pi} - a_j + 2N\ell\right) + g_j\left(-\frac{N\theta}{\pi} - a_j + 2N\ell\right) \right]. \tag{5.3}$$

The series converges uniformly by (2.3). Its Fourier coefficients vanish at all frequencies k for which

$$\frac{\pi|k|}{N} > \pi - 4\gamma,$$

by the Paley–Wiener theorem. It is therefore an even real trigonometric polynomial of degree less than N, and hence an algebraic polynomial of degree less than N in $x = \cos\theta$.

Because I^θ stays away from 0 and π, the reflected centers stay a fixed angular distance from $[0,\pi]$, modulo 2π. Their contribution, and that of every nonzero period of the unreflected term, is $O(N^{-3})$ uniformly on $[0,\pi]$, by (2.3). Summing these errors over $O(N)$ retained boxes gives $o(1)$.

By (2.3) and (5.2), the sum of the principal tails from all other retained boxes is less than $1/10$. For each retained box let

$$B_{N,j}(x) = \Phi_{N,j}(\arccos x).$$

If a node $z = \cos\theta$ lies in the j-th retained box, the principal term of $B_{N,j}(z)$ has absolute value at most one by (2.1); if it lies outside that box, the same term is controlled by (2.3). The reflected and nonzero-period terms contribute only $o(1)$ after summation. Therefore, for every choice of signs $\varepsilon_j \in \{-1, 1\}$ and all sufficiently large N,

$$\max_{z \in Y} \left| \sum_j \varepsilon_j B_{N,j}(z) \right| \le 2. \tag{5.4}$$

On the other hand, each retained box contains a node-free interval of a fixed positive length in the v-coordinate on which its principal function is at least $c' \log\rho > 10H$. By the choice of D, on this interval its own principal term dominates all the other terms. After dividing the sum in (5.4) by two, each retained box therefore contains a node-free interval on which the resulting polynomial has absolute value greater than $2H$, with sign prescribed by ε_j. Order these peak intervals from left to right and choose the signs so that

$$\operatorname{sgn}(b/P_Y)|_{J_j} = (-1)^j.$$

The polynomial b has degree less than N, and (5.4) gives $|b(y)| \le 1$ for every $y \in Y$. Setting the interpolation data to $v_y = b(y)$, any interpolant $p \in \Pi_{N+d}$ has the form

$$p = b + P_Y q, \qquad \deg q \le d.$$

There are $c_{\rho,D} N |I^\theta|$ retained peak intervals, whereas $d = o(N)$. Lemma 3 therefore leaves a fixed positive fraction of them on which $|p| > H$ everywhere.

Each peak interval has length bounded below by c/N in the θ-coordinate. For sufficiently small I about any fixed point of $(-1,1)$, the maximum and minimum of $\sin\theta$ on I^θ have ratio at most two. Hence the total x-length of the surviving intervals is bounded below by a fixed multiple of $|I|$, and we obtain

$$|\{x \in I : |p(x)| > H\}| \geq c_H |I| \tag{5.5}$$

for all interpolants of degree $N + o(N)$. The constant depends only on H, provided (5.1) and this ratio bound hold. This argument does not require the number of nodes in I to tend to infinity.

6 Local intervals for an arbitrary node array

Pass to a subsequence, with indices still denoted by n, such that

$$\nu_n = \frac{1}{n} \sum_{x \in X_n} \delta_x \implies \nu. \tag{6.1}$$

Only this subsequence will be used in the construction. Removing the nodes belonging to any fixed finite set does not alter the weak limit. Put

$$V = V_\nu, \qquad m_0 = \inf_{\operatorname{supp} \nu} V.$$

The function V belongs to $L^1[-1,1]$, and $V \geq m_0$ throughout $[-1,1]$. To see the latter assertion off the support, note that V is concave on each bounded complementary interval and has the corresponding endpoint limits. On the two exterior intervals it is monotone toward the support.

6.1 Points above the minimum potential

Fix a finite set $S \subset [-1,1]$ of nodes whose values have already been specified, and put

$$Y_n = X_n \setminus S, \qquad P_n = P_{Y_n}.$$

For $z \in \operatorname{supp} \nu$, choose $z_n \in Y_n$ with $z_n \to z$. Truncation of the logarithmic kernel gives

$$\limsup_{n \to \infty} \frac{1}{n} \log |P_n'(z_n)| \leq V(z), \qquad \frac{1}{n} \log |P_n| \longrightarrow V \quad \text{in } L^1[-1,1]. \tag{6.2}$$

For the first assertion, truncate $\log |x - y|$ from below at $-A$. The contribution of the omitted diagonal term is A/n; one first takes the weak limit and then lets $A \to \infty$. For the second, use the continuity of

$$y \longmapsto (x \longmapsto \log |x - y|)$$

as a map from $[-1,1]$ into $L^1[-1,1]$.

Suppose first that $V(z) > -\infty$, and let

$$A_{z,\eta} = \{x : V(x) > V(z) + 3\eta\}, \qquad \eta > 0.$$

Outside a subset of $A_{z,\eta}$ whose measure tends to zero, (6.2) implies

$$|\ell_n(x)| \geq e^{\eta n}, \qquad \ell_n(x) = \frac{P_n(x)}{(x - z_n) P_n'(z_n)}. \tag{6.3}$$

Specify the value one at z_n, and zero at all other nodes of Y_n. Every polynomial of degree at most $n + r_n$ satisfying these data can be written as

$$p = \ell_n T, \qquad T(z_n) = 1, \qquad \deg T \leq r_n + \#S + 1 = o(n). \tag{6.4}$$

If $|p| \le H$ on a subset of $A_{z,\eta}$ of fixed positive measure, (6.3) and Remez's inequality[2] yield

$$|T(z_n)| \le He^{-\eta n} C^{r_n + \#S + 1} < 1$$

for all sufficiently large n, a contradiction. Here C depends only on the fixed measure lower bound. Thus the measure of the low set in $A_{z,\eta}$ tends to zero uniformly over all the polynomials in (6.4). At every density point of $A_{z,\eta}$, each sufficiently small fixed interval therefore admits data forcing large values on at least half its length.

Choose countably many z for which $V(z)$ decreases to m_0, and take positive rational η. The sets $A_{z,\eta}$ cover $\{V > m_0\}$ when the chosen source potentials are finite. If some $V(z) = -\infty$, the first assertion of (6.2) instead gives

$$\frac{1}{n} \log |P_n'(z_n)| \longrightarrow -\infty.$$

On each set $\{V \ge -A\}$, the same argument gives (6.3) for any fixed positive exponential rate, outside a set of vanishing measure. Letting $A \to \infty$ covers almost all of $[-1, 1]$. In particular, if $m_0 = -\infty$, this subsection supplies local intervals at almost every point.

6.2 The minimum-potential set

Suppose that $m_0 > -\infty$, and let

$$E = \{x : V(x) = m_0\}.$$

By the differentiation theorem for measures, ν has a finite Lebesgue density at almost every point. At points where this density is zero, all sufficiently small intervals satisfy the strict limiting form of (5.1). Fixing such an interval before taking $n \to \infty$, and choosing endpoints that are not atoms of ν, allows Section 5 to be applied.

We isolate the local field arising at the remaining points.

Lemma 7 (Local outer field)**.** *Let x_0 be a density point of E and a Lebesgue point of the absolutely continuous part of ν, with finite positive density w, and suppose that the singular part of ν has density zero at x_0. Put*

$$I_h = (x_0 - h, x_0 + h), \qquad a_h = \nu(I_h),$$

and let μ_h be the image of $\nu|_{I_h}/a_h$ under $x \mapsto (x - x_0)/h$. Then, as $h \downarrow 0$,

$$a_h \sim 2wh, \qquad \mu_h \Longrightarrow \frac{du}{2},$$

and

$$V_{\mu_h} \longrightarrow Q - 1 \quad \textit{in } L^1(-1, 1). \tag{6.5}$$

Define the normalized outer potential[3]

$$A_h(u) = -\frac{1}{a_h} \int_{\mathbb{R} \setminus I_h} \log |x_0 + hu - y| \, d\nu(y).$$

Then

$$A_h(u) - A_h(0) \longrightarrow Q(u) \quad \textit{locally uniformly on } (-1, 1). \tag{6.6}$$

[2]An earlier version of this argument was developed with GPT-5.6.

[3]The local external-field model was identified in an earlier GPT-5.6-assisted round.

Moreover, for every compact $K \Subset (-1,1)$, the functions $A_h - A_h(0)$ admit analytic continuations $\widetilde{A}_h$ to a fixed complex neighborhood of K, and

$$\widetilde{A}_h \longrightarrow Q$$

locally uniformly there.

Proof. The differentiation theorem gives $a_h \sim 2wh$ and $\mu_h \Rightarrow du/2$. The continuity of

$$y \longmapsto (x \longmapsto \log|x-y|)$$

from $[-1,1]$ into $L^1[-1,1]$ gives (6.5), since the logarithmic potential of $du/2$ is $Q-1$.

For u corresponding to $E \cap I_h$, the equality $V = m_0$ gives

$$A_h(u) + \frac{m_0}{a_h} - \log h = V_{\mu_h}(u). \tag{6.7}$$

The set of such u occupies a fraction tending to one of $(-1,1)$, and the left side of (6.7) is convex. Thus (6.5) and (6.7) imply convergence in measure to $Q-1$. Finite convex functions that converge in measure to a continuous finite convex function converge uniformly on compact subintervals: convergence points on either side of a compact interval bound the secant slopes, giving equicontinuity. This proves (6.6).

Fix $K \Subset (-1,1)$, and choose a compact interval K_1 whose interior contains $K \cup \{0\}$. Let U be a simply connected complex neighborhood of K_1 whose closure stays a positive distance from $(-\infty,-1] \cup [1,\infty)$. Writing $\xi = (y - x_0)/h$, one has $|\xi| \geq 1$ for $y \notin I_h$. Define $\widetilde{A}_h$ on U by $\widetilde{A}_h(0) = 0$ and

$$\widetilde{A}'_h(z) = -\frac{h}{a_h} \int_{\mathbb{R}\setminus I_h} \frac{d\nu(y)}{x_0 + hz - y}.$$

For real $u \in K_1$, integration along the real segment gives $\widetilde{A}_h(u) = A_h(u) - A_h(0)$, and

$$\widetilde{A}''_h(z) = \frac{1}{a_h} \int_{\mathbb{R}\setminus I_h} \frac{d\nu(y)}{(z-\xi)^2}.$$

For sufficiently small fixed $\ell > 0$, the kernels $(u-\xi)^{-2}$ are uniformly comparable on $[u-\ell, u+\ell]$, uniformly for $u \in K_1$. Hence

$$A''_h(u) \leq C_\ell \int_{u-\ell}^{u+\ell} A''_h(v)\, dv = C_\ell \big(A'_h(u+\ell) - A'_h(u-\ell)\big).$$

The local uniform convergence in (6.6) and convexity give uniform interior bounds for A'_h, hence for A''_h on K_1. In a sufficiently thin complex neighborhood of K_1,

$$|z-\xi|^{-2} \leq |\operatorname{Re} z - \xi|^{-2},$$

so $\widetilde{A}''_h$ is uniformly bounded there as well. Together with the normalization at zero and the interior first-derivative bound, this gives a normal family. Every subsequential limit agrees with Q on the real interval by (6.6), and therefore equals Q by the identity theorem. □

Lemma 8 (Discrete outer-field approximation)**.** *Fix $h > 0$ such that the endpoints of I_h and the preimages of the endpoints of A are not atoms of ν. Let S be a fixed finite set of nodes and put*

$$Y_n = X_n \setminus S, \qquad s_n = \#(Y_n \cap I_h).$$

Then

$$\frac{s_n}{n} \longrightarrow a_h, \qquad \frac{\#\{y \in Y_n : (y - x_0)/h \in A\}}{s_n} \longrightarrow \mu_h(A). \tag{6.8}$$

Let $\mathcal{Q}_n$ be the normalized analytic continuation of the discrete field (4.1), with $\mathcal{Q}_n(0) = 0$. On every fixed complex neighborhood compactly contained in the domain of $\widetilde{A}_h$,

$$\mathcal{Q}_n \longrightarrow \widetilde{A}_h$$

locally uniformly. If $N_n = \#Y_n$, then the degree bound $n + r_n$ is $N_n + d_n$, where

$$d_n = r_n + \#(X_n \cap S) = o(s_n). \tag{6.9}$$

Proof. The two limits in (6.8) follow from weak convergence and the choice of the interval endpoints. Moreover,

$$\mathcal{Q}_n'(z) = -\frac{h}{s_n} \sum_{y \in Y_n \setminus I_h} \frac{1}{x_0 + hz - y}.$$

On a fixed complex neighborhood as above, the summands are uniformly bounded and equicontinuous as functions of the outer node. Since $n/s_n \to 1/a_h$, weak convergence of the empirical measures gives $\mathcal{Q}_n' \to \widetilde{A}_h'$ locally uniformly. Integrating from zero gives the asserted convergence of the fields. Finally, $r_n = o(n)$ and $s_n/n \to a_h > 0$ give (6.9). □

Now let x_0 satisfy the hypotheses of Lemma 7; these conditions hold at almost every point of E where the density of ν is positive. Apply that lemma with $K = [-b, b]$. Choose h so small that $\widetilde{A}_h$ belongs to the analytic neighborhood required in Proposition 6 and

$$\mu_h(A) < \frac{1}{4} + \frac{\epsilon_\rho}{2}.$$

This is possible because $\mu_h \Rightarrow du/2$ and $(du/2)(A) = 1/4$. We may also choose h so that the interval endpoints required in Lemma 8 are not atoms of ν. That lemma then gives, for all sufficiently large n, (4.2), the required convergence of the discrete external field, and an excess degree $d_n = o(s_n)$. Therefore Proposition 6 applies.

Combining the two subsections, and decreasing c_H if necessary, we have proved the following local conclusion. For almost every $x \in (-1, 1)$, there are arbitrarily small intervals I containing x such that, after any fixed finite set of nodes has been removed, every sufficiently late row of the subsequence admits data bounded by one on the remaining nodes and forcing

$$|\{u \in I : |p_n(u)| > H\}| \geq c_H |I|$$

for every interpolant of degree at most $n + r_n$. The constant $c_H > 0$ depends only on H.

7 A finite construction for common low sets

Proposition 9. *Fix $H > 1$, $\delta > 0$, and $N_0 \geq 1$. There are finitely many indices $n_1, \ldots, n_k \geq N_0$ and a function $g \in C([-1, 1]; \mathbb{R})$, with $\|g\|_\infty \leq 1$, such that every choice of polynomials*

$$p_j \in \Pi_{n_j + r_{n_j}}, \qquad p_j = g \ \text{on}\ X_{n_j},$$

satisfies

$$\left| \bigcap_{j=1}^{k} \{x : |p_j(x)| \leq H\} \right| < \delta. \tag{7.1}$$

Proof. For any interval $J \subset (-1,1)$, the intervals furnished by Section 6 form a Vitali cover of almost all of J. Choose finitely many disjoint such intervals whose total length is at least $|J|/2$. For each interval, choose a sufficiently late new row of the fixed subsequence, assigning values only at nodes whose values have not yet been specified. For all choices of the resulting interpolating polynomials, the union of their sets of values exceeding H occupies at least

$$c_*|J|, \qquad c_* = \frac{c_H}{2} > 0, \tag{7.2}$$

inside J.

To justify the compatibility of these assignments, let S be the finite set of previously assigned nodes. At a new row n, use only $Y_n = X_n \setminus S$ in the local construction. If b is its polynomial of degree at most $\#Y_n - 1$, every polynomial interpolating all the assigned values on X_n has the form

$$p = b + P_{Y_n} q, \qquad \deg q \le r_n + \#(X_n \cap S) \le r_n + \#S.$$

Since $\#S$ is fixed before n is chosen, the additional degree is still $o(n)$, so the local conclusion remains applicable.

Assume $0 < \delta \le 2$, since larger δ is immediate. Suppose that finitely many rows have already been chosen, with degree bounds $d_1, \dots, d_k$. For every admissible choice of their polynomials, the common low set

$$L = \bigcap_{j=1}^{k} \{x : |p_j(x)| \le H\}$$

has at most $2 + 2\sum_j d_j$ boundary points in $[-1,1]$. A constant polynomial equal to H or $-H$ introduces no additional boundary points. Choose a fixed partition of $[-1,1]$ into sufficiently short intervals that the cells meeting these boundary points have total length less than $\delta/4$. The same partition works for every admissible choice of the polynomials, since the boundary bound depends only on their degrees.

In each cell, carry out the construction giving (7.2), using new rows and assigning only new node values. If $|L| \ge \delta$, the cells contained entirely in L have total length at least

$$|L| - \delta/4 \ge 3|L|/4.$$

For the new common low set this gives

$$|L_{\text{new}}| \le (1 - 3c_*/4)|L| \qquad \text{whenever } |L| \ge \delta. \tag{7.3}$$

Starting from $L = [-1,1]$, perform a finite number q of batches, where

$$2(1 - 3c_*/4)^q < \delta.$$

At each batch the new grid is chosen using the degree bounds of the rows already selected. Equation (7.3) shows that the final common low set has measure less than δ, uniformly over all admissible polynomial choices. All rows may be chosen beyond N_0.

The assigned values define a function on a finite union of nodes, with absolute value at most one. Extend it by linear interpolation between consecutive nodes and by constants to the endpoints of $[-1,1]$. The resulting continuous function g has $\|g\|_\infty \le 1$ and satisfies (7.1). $\square$

8 Proof of Theorem 1

Work in the real Banach space $C([-1,1];\mathbb{R})$. For $H>0$ and positive integers N,k, let $\mathcal{O}_{H,N,k}$ consist of the functions f for which there is a finite set $S\subset\{N,N+1,\dots\}$ such that every choice

$$p_n\in\Pi_{n+r_n},\qquad p_n=f \text{ on } X_n\quad (n\in S),$$

satisfies

$$\left|\bigcap_{n\in S}\{x:|p_n(x)|\le H\}\right|<\frac{1}{k}.$$

Lemma 10. *If $H>M>0$, then $\mathcal{O}_{H,N,k}\subset \operatorname{int}\mathcal{O}_{M,N,k}$.*

Proof. Let $f\in\mathcal{O}_{H,N,k}$, with finite set S as above. For each $n\in S$, let L_n be the Lagrange interpolation operator on X_n, and write $\ell_{n,\xi}$ for its fundamental polynomials. Put

$$C=1+\sum_{n\in S}\sum_{\xi\in X_n}\|\ell_{n,\xi}\|_\infty .$$

If $\|g-f\|_\infty<(H-M)/C$ and p_n interpolates g, then $q_n=p_n-L_n(g-f)$ interpolates f with the same degree bound. Moreover,

$$\|L_n(g-f)\|_\infty<H-M,\qquad \{|p_n|\le M\}\subset\{|q_n|\le H\}.$$

The defining bound for f therefore gives the defining bound for g at height M. Thus the ball of radius $(H-M)/C$ about f lies in $\mathcal{O}_{M,N,k}$. □

Lemma 11. *For every positive integer triple M,N,k, the open set $\operatorname{int}\mathcal{O}_{M,N,k}$ is dense.*

Proof. Let an open ball in $C([-1,1];\mathbb{R})$ be given. By polynomial approximation, it contains a polynomial h. Choose $\varepsilon>0$ so that $h+\varepsilon g$ belongs to the ball whenever $\|g\|_\infty\le 1$. Apply Proposition 9 with

$$H>\max\left\{1,\frac{M+1+\|h\|_\infty}{\varepsilon}\right\},\qquad \delta=\frac{1}{k},$$

requiring all selected indices to be at least N and large enough that $n+r_n\ge\deg h$. Let g be the resulting function and set $f=h+\varepsilon g$. For any admissible interpolant p_n of f on a selected row,

$$q_n=\frac{p_n-h}{\varepsilon}$$

is an admissible interpolant of g, and $|p_n|\le M+1$ implies $|q_n|<H$. Hence $f\in\mathcal{O}_{M+1,N,k}$. Lemma 10 puts f in $\operatorname{int}\mathcal{O}_{M,N,k}$, as required. □

Proof of Theorem 1. By Lemma 11 and the Baire category theorem, choose

$$f\in\bigcap_{M,N,k\ge1}\operatorname{int}\mathcal{O}_{M,N,k}.$$

Fix any admissible interpolation sequence (p_n). For every M,N,k, the common tail low set

$$E_{M,N}=\bigcap_{n\ge N}\{x:|p_n(x)|\le M\}$$

is contained in the finite common low set supplied by $f\in\mathcal{O}_{M,N,k}$. Thus $|E_{M,N}|<1/k$ for every k, and $|E_{M,N}|=0$. Outside the null set $\bigcup_{M,N\ge1}E_{M,N}$, every tail of $(|p_n(x)|)$ exceeds every positive integer threshold. This proves (1.1). □

Use of generative AI

AI tools were used substantially in the development of this work. An earlier round of work with GPT-5.6 developed several ingredients that contributed to the final proof, including the sign-change mechanism used to preserve intervals of large values against low-degree corrections, the Remez argument on regions of higher logarithmic potential, and the local external-field model near the minimum-potential set.

Building on notes from that round, GPT-6 Astra connected the external-field model to weighted polynomial spaces through Christoffel–Darboux kernel asymptotics and localization, and developed the finite construction used to combine data from multiple rows.

The Lean formalization and the finite-perturbation category argument in Section 8 were developed with OpenAI Codex (GPT-6).

The author checked the final arguments and the external results on which they depend against the cited sources, and is solely responsible for the mathematical content of the paper.

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